

Credence: A Bayesian Decision-Theoretic Framework for LLM Agent Tool Selection

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June 5, 2026

Abstract

Current LLM agent frameworks select tools through prompting heuristics without any formal mechanism for evaluating whether a tool query is worth its cost. We introduce CREDENCE, a Bayesian decision-theoretic framework that derives tool selection from first principles: agents maintain conjugate beliefs about tool reliability, update them from observed outcomes, and select queries by maximising expected utility via value-of-information calculations. The framework is grounded in five axioms—Bayesian conditioning, expected utility integration, EU maximisation, a Solomonoff complexity prior, and a cooperative inverse reinforcement learning (CIRL) alignment commitment—from which all agent behaviour is derived rather than engineered.

We study *when* this pays, on a controlled benchmark of category-dependent tool reliability, under two conditions: question categories *given* (oracle) and *inferred* by a Gaussian Naive-Bayes classifier on offline sentence embeddings (leave-one-out accuracy 0.78). Our central finding is a *contingency law*: the value of exploration is contingent on category-attribution quality. A horizon-aware VOI policy—valuing information over the remaining question horizon by pure expected-utility maximisation, with exploration emergent rather than engineered—*beats* an optimism-under-uncertainty greedy rule by +27 when categories are given, but *loses* once they are inferred, because attribution noise corrupts the per-category reliability signal exploration depends on. The crossover is robust across every credit-assignment rule we test. Independently of which action policy wins, the Bayesian substrate owns the cost-efficient frontier: the myopic-VOI agent dominates a free local LLM (Llama 3.1 8B) in both tool cost and gross reward, and only a paid commercial model (Claude Haiku 4.5—and only its fast, low-cost tier, 97.5% accuracy at \$3.24 per 1000 questions) beats the family’s 56–64% at real dollar cost. We also surface an *accuracy paradox*—the query-all agents reach the highest non-LLM accuracy (64–69%) yet the worst scores (−67.0, −24.5), as tool costs destroy the margin—and confirm reliability learning as the dominant component. We frame these as laws about *when* principled tool selection pays, not a claim that value-of-information is universally optimal.

1 Introduction

The rapid deployment of large language model (LLM) agents—systems that use LLMs to select and invoke external tools—has created a curious gap in AI practice. While the perception capabilities of these systems have improved dramatically through scaling [Sutton, 2019], the *decision-making* layer remains largely unprincipled. Frameworks such as LangChain [Chase, 2022] and LlamaIndex implement tool selection through prompting: the LLM is instructed to reason about which tools to

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use, with no formal mechanism for quantifying uncertainty, evaluating marginal information value, or trading off query costs against expected gains.

This paper argues that the tool selection problem is fundamentally a *decision-theoretic* problem, not a perception problem, and that treating it as such yields both theoretical clarity and practical improvements. Decision theory does not compete with deep learning for the same job—it answers a different question entirely: given uncertainty about tool reliability and finite query budgets, *what should the agent do?*

We introduce CREDENCE, a framework built on five axioms from which all agent behaviour is derived. The key insight is that tool queries are information-gathering actions whose value can be computed exactly: the *value of information* (VOI) of querying a tool is the expected utility of observing the tool’s response and then acting optimally, minus the expected utility of acting on current beliefs alone. A query is worth making if and only if its VOI exceeds its cost. This is not a heuristic—it is a theorem of Bayesian decision theory [Raiffa and Schlaifer, 1961, DeGroot, 1984].

Contributions. (1) We present a principled framework for LLM tool selection grounded in Bayesian decision theory and CIRL, requiring no LLM calls for the decision-making layer itself. (2) Our headline empirical finding is a *contingency law*: the value of exploration in category-conditioned tool selection is contingent on category-attribution quality. With *given* categories a horizon-aware VOI policy beats an optimism-greedy rule by +27; with *inferred* categories (a 0.78-accuracy classifier) it loses, because attribution noise corrupts the per-category reliability signal—a crossover robust across every credit-assignment rule we test. (3) We introduce *horizon-aware VOI*, a constitution-clean exploration mechanism: it values information over the remaining question horizon by pure expected-utility maximisation, with exploration emergent (no ϵ -greedy, no exploration bonus), and a depth-1 lookahead already captures the full gain—no deep search required. (4) We characterise the *cost-efficient frontier*: the Bayesian family owns the cheap regime in both conditions—the myopic-VOI agent dominates a free local LLM in tool cost *and* gross reward, and only a paid commercial model beats it—and we surface, as a secondary observation, the *accuracy paradox* (highest accuracy coincides with worst score when tool costs are non-zero). (5) As a subordinate structural result we report a credit-rule $\times$ query-strategy interaction, and we disclose that the deployed credit rule is an acknowledged approximation to the proper posterior-weighted update.

Connection to prior work. The present framework builds on two lines of research. First, the theory of *embodied agent design* due to Ay [2015], which formalises the separation between an agent’s controller (brain) and its sensorimotor apparatus (body and environment), and argues that control architectures should be *sufficient* for expressing desired behaviours while remaining *concise*—the principle of cheap design. In CREDENCE, the “brain” is the Bayesian decision layer; the “body” is the LLM and external tools. The LLM is treated as a *prosthetic*—a noisy sensor whose reliability is learned, not a decision-maker.

Second, the author’s prior work on *dynamic staged trees* [Freeman and Smith, 2011b] and *chain event graph* (CEG) model selection [Freeman and Smith, 2011a], which demonstrated that Bayesian structure learning with conjugate updates can handle non-stationary discrete processes through multi-process switching. The Beta-Bernoulli reliability model in CREDENCE is a specialisation of the Dirichlet-categorical conjugate framework used for CEG stage learning.

2 Related Work

2.1 Principled Frameworks for LLM Agents

The application of Bayesian decision theory to agent design has a long history [Russell and Wefald, 1991, Boutilier et al., 1999, Kaelbling et al., 1998], yet modern LLM agent frameworks have largely ignored these foundations. ReAct [Yao et al., 2023], Reflexion [Shinn et al., 2023], and LATS [Zhou et al., 2024] delegate both perception and decision-making to the LLM (the latter via MCTS-style planning); Toolformer [Schick et al., 2023] fine-tunes models to decide when to call APIs. These approaches conflate two fundamentally different capabilities: extracting information from tool outputs (perception) and choosing which tools to invoke (decision-making). Wang et al. [2024] survey over 100 such systems; none maintains explicit beliefs about tool reliability or computes value of information.

Three recent works apply principled methods to LLM agents. **RAFA** [Liu et al., 2024] is the most formally rigorous: it casts agent reasoning as a Bayesian adaptive MDP and proves a $\sqrt{T}$ regret bound. However, RAFA addresses planning in unknown environments, not per-tool reliability or VOI for individual tool calls. **DeLLMa** [Liu et al., 2025] applies expected utility maximisation to LLM decisions via multi-step inference—the closest in spirit to CREDENCE’s EU framework—but lacks belief updating, VOI computation, and tool reliability tracking. **MACLA** [Forouzandeh et al., 2026] maintains Beta posteriors over procedure success rates, and its ablation confirms Bayesian selection as the single most impactful component. But MACLA selects multi-step procedures, not individual tools, and does not compute VOI.

To our knowledge, no published work combines per-tool Bayesian belief tracking, value-of-information computation, and expected utility maximisation for LLM tool selection.

2.2 Cost-Aware Agent Approaches

Cost-aware approaches to LLM tool use span five tiers of increasing formality: (1) *prompt-based*—expose budget or cost information in the prompt; (2) *RL reward shaping*—penalise tool calls in the reward signal; (3) *planning and optimisation*—pre-plan tool sequences under budget constraints, as in BTP [Zheng et al., 2024]; (4) *sequential decision-making*—online planning with cost anticipation, most rigorously formalised by INTENT [Liu et al., 2026]; and (5) *decision-theoretic*—computing VOI before each tool call, the approach taken by CREDENCE.

Kapoor et al. [2025] demonstrate that agent evaluation without cost accounting is misleading—accuracy alone fails to capture real-world value. Wang et al. [2024] show that budget-aware reasoning strategies outperform unconstrained ones, and Erol et al. [2025] propose an economic framework for model evaluation that accounts for inference cost. Wang et al. [2025] argue that agents should invoke tools “only when epistemically necessary,” defining a knowledge boundary concept but providing no computational mechanism. A Bayesian VOI framework provides exactly the mechanism they call for.

Our benchmark results (§5) illustrate the cost-performance frontier: a local LLM with cost-awareness prompts and tool-calling still scores barely half the best zero-cost Bayesian agent, whilst a commercial LLM outperforms it only at non-trivial API cost. Natural language cannot express the quantitative tradeoff between information gain and query cost that VOI captures analytically. De Sabbata et al. [2024] reach a complementary conclusion from the metareasoning perspective: LLMs benefit from explicit computational cost-benefit analysis rather than implicit prompt-level reasoning.

2.3 Theoretical Foundations

Embodied agent design. Ay [2015] formalises the sensorimotor loop as a causal model with Markov kernels and shows that an agent’s controller need not be a universal approximator—it suffices to match the behavioural repertoire afforded by its embodiment. This *cheap design* principle [Pfeifer and Bongard, 2006, Montúfar et al., 2015] motivates CREDENCE’s architecture: the Bayesian decision layer is far simpler than the LLM it coordinates, yet sufficient for effective tool selection given the embodiment constraints (available tools and their reliability profiles).

Graphical models. Chain event graphs [Smith and Anderson, 2008, Freeman and Smith, 2011a, Thwaites et al., 2009] generalise Bayesian networks to handle context-specific independence with conjugate model selection via product Dirichlet priors. Dynamic staged trees [Freeman and Smith, 2011b] extend this to non-stationary time series through multi-process switching models. The per-tool, per-category Beta-Bernoulli reliability model in CREDENCE is a specialisation of the product Dirichlet conjugate framework developed for chain event graph model selection [Freeman and Smith, 2011a], where Freeman and Smith [2011a, Theorem 18] showed that path and floret independence assumptions characterise the Dirichlet prior on staged tree parameters. Each tool-category pair in CREDENCE corresponds to a floret in the CEG sense, with the correct/incorrect outcome defining a two-edge floret whose natural conjugate prior is Beta.

Value of information. VOI originates with Howard [1966] and is developed for bounded agents by Russell and Wefald [1991]. Bayesian experimental design [Chaloner and Verdinelli, 1995] formalises optimal information gathering under cost constraints. Dong et al. [2026] recently extend VOI to human-agent communication, but apply it to dialogue acts rather than tool selection. CREDENCE appears to be the first system to compute VOI for individual tool calls in an LLM agent.

Value alignment. Cooperative inverse reinforcement learning [Hadfield-Menell et al., 2016] treats human-robot interaction as a cooperative game in which the robot infers human preferences from observed behaviour. CREDENCE’s treatment of the LLM as a noisy sensor—whose outputs are observations with uncertainty, not commands—embodies a similar principle: the decision-theoretic layer maintains its own beliefs and defers to the human’s utility function rather than to the LLM’s token probabilities.

3 The CREDENCE Framework

3.1 Axioms

CREDENCE is built on five axioms. These are non-negotiable foundations from which all behaviour is derived.

Axiom 1 (Conditioning). *Beliefs are updated by Bayes’ rule. This is the unique update rule for which updating and computing the predictive commute, and the unique rule that cannot be Dutch-booked across time (diachronic coherence).*

Axiom 2 (Integration). *The expected value of a quantity under a belief is computed by integrating over the belief distribution.*

Axiom 3 (Optimisation). *Actions are selected to maximise expected utility under the current belief. Savage’s representation theorem [Savage, 1954] establishes that any agent whose preferences satisfy*

basic rationality axioms acts as if maximising expected utility. The complete class theorem [Wald, 1950, Berger, 1985] further establishes that Bayesian decision rules are the only admissible ones: any non-Bayesian rule is dominated by some Bayesian rule.

Axiom 4 (Complexity prior). *The prior over hypotheses is weighted by description length: $2^{-|h|}$ for hypothesis h . This is Solomonoff’s universal prior [Solomonoff, 1964]—the unique prior that dominates all computable priors up to a constant.*

Axiom 5 (Alignment). *The agent’s utility equals the user’s utility under preference uncertainty: $U(a, s, b) = \mathbb{E}_{\theta \sim b}[u_{\theta}(s, a)]$, where θ parameterises the user’s preferences and b is the agent’s belief over θ . This is the cooperative inverse reinforcement learning (CIRL) framework of Hadfield-Menell et al. [2016]: the agent and user are in a cooperative game where the agent’s reward is the user’s reward, but the agent does not know the user’s reward function and must infer it from interaction.*

Scope of this paper. The experiments here exercise Axioms 1–3 (conditioning, integration, EU-maximisation), which fully determine the tool-selection behaviour we evaluate. The complexity prior (Axiom 4) and the CIRL alignment commitment (Axiom 5) are framework-level commitments developed in the companion work; they do not act on the present benchmark, where the utility function is fixed and known and the hypothesis space is enumerated. In particular, the Beta(1, 1) reliability prior introduced below is a per-tool conjugate object over a *reliability parameter*—distinct from, and not in tension with, the Axiom-4 description-length prior over *hypotheses*.

3.2 Architecture: Brain, Body, Environment

Following Ay [2015], we decompose the agent into three components:

- **Brain** (controller $\mathcal{C}$): The Bayesian decision layer. Maintains beliefs about tool reliability, computes expected utilities, selects actions. Implemented in ~ 200 lines of Julia. Contains no LLM.
- **Body** (actuators and sensors): The available tools, including the LLM itself. These are *noisy channels* in the information-theoretic sense—Markov kernels $\beta : \mathcal{W} \rightarrow \Delta_{\mathcal{S}}$ from world states to sensor states, following the notation of Ay [2015]. The LLM is a prosthetic: a powerful but unreliable sensor whose reliability profile is learned, not assumed.
- **Environment** (world $\mathcal{W}$): The questions, their categories, and the ground-truth answers. The agent does not observe the environment directly—only through the noisy responses of its tools.

This separation instantiates the principle of cheap design: the brain is deliberately simple because the body (LLM + tools) provides the perceptual complexity. The brain’s job is not to understand language or reason about content—it is to decide *which tool to query and when to stop*.

3.3 Belief Model

For each tool $t \in \{1, \dots, T\}$ and question category $c \in \{1, \dots, C\}$, the agent maintains a Beta-distributed belief over the tool’s reliability:

$$p_{t,c} \sim \text{Beta}(\alpha_{t,c}, \beta_{t,c}). \quad (1)$$

The conjugate Beta-Bernoulli model means that after observing a correct response from tool t on a category- c question, the update is:

$$(\alpha_{t,c}, \beta_{t,c}) \mapsto (\alpha_{t,c} + 1, \beta_{t,c}), \quad (2)$$

and after an incorrect response:

$$(\alpha_{t,c}, \beta_{t,c}) \mapsto (\alpha_{t,c}, \beta_{t,c} + 1). \quad (3)$$

Initialisation. Priors are set to Beta(1,1) (uniform) for each tool-category pair. This encodes maximum ignorance about reliability, placing the prior mean at $1/2$ with high variance. The choice of $1/N$ (where N is the number of tools) for prior success probability, rather than 0.5, was considered and rejected: the Beta(1,1) prior is the unique non-informative conjugate prior and makes no assumption about relative tool quality.

Non-stationarity. Tool reliability genuinely drifts in deployment—commercial LLM behaviour, for instance, changes measurably over time [Chen et al., 2023]. The core framework handles such distributional drift through standard Beta-Bernoulli updating: as a tool’s reliability degrades, negative evidence accumulates and the posterior shifts downward naturally. No explicit change-detection mechanism is required—the posterior adapts continuously, with the rate of adaptation governed by the effective sample size (the sum $\alpha + \beta$).

Non-stationarity extensions. The Beta-Bernoulli model adapts continuously to changing tool reliability through standard posterior updating: as negative evidence accumulates, the posterior shifts downward, governed by the effective sample size ($\alpha + \beta$). For abrupt regime changes, principled extensions include Bayesian online changepoint detection [Adams and MacKay, 2007] and power steady models [Freeman and Smith, 2011b], which we defer to future work.

3.4 Decision Rule: Value of Information

The agent’s decision at each step is: should I query another tool, or should I act (submit an answer or abstain) based on current beliefs?

Let b denote the current belief state (the collection of all Beta parameters). Let r_t denote the cost of querying tool t . Let $v_{\text{correct}} = 10$, $v_{\text{wrong}} = -5$, $v_{\text{abstain}} = 0$ be the payoffs.

Expected utility of acting now. Given current beliefs b , if the agent has gathered responses $\{(t_i, x_i)\}_{i=1}^k$ from k tool queries, it computes a posterior probability p_{correct} of the most-supported answer being correct (by aggregating the reliability-weighted evidence). The expected utility of submitting is:

$$\text{EU}_{\text{submit}}(b) = p_{\text{correct}} \cdot v_{\text{correct}} + (1 - p_{\text{correct}}) \cdot v_{\text{wrong}}. \quad (4)$$

The expected utility of abstaining is $\text{EU}_{\text{abstain}} = 0$. The agent submits if $\text{EU}_{\text{submit}} > 0$, i.e., if $p_{\text{correct}} > 1/3$ (the break-even threshold given the payoff structure).

The best “act now” utility is:

$$\text{EU}^*(b) = \max(\text{EU}_{\text{submit}}(b), 0). \quad (5)$$

Value of information. The VOI of querying tool t is:

$$\text{VOI}(t \mid b) = \mathbb{E}_{x \sim p_t}[\text{EU}^*(b \mid x)] - \text{EU}^*(b), \quad (6)$$

where the expectation is over the possible responses x of tool t , weighted by the agent’s current belief about tool t ’s reliability. This is the expected improvement in decision quality from observing tool t ’s response.

A query is made if and only if:

$$\text{VOI}(t \mid b) > r_t \quad \text{for some tool } t. \quad (7)$$

When multiple tools satisfy this criterion, the agent selects the one with the highest net VOI: $\text{argmax}_t[\text{VOI}(t \mid b) - r_t]$.

Proposition 1. *Under the Beta-Bernoulli model, $\text{VOI}(t \mid b)$ is non-increasing in the number of prior observations. That is, the more the agent knows about a tool, the less informative an additional query becomes.*

Proof. As $\alpha_{t,c} + \beta_{t,c} \rightarrow \infty$, the posterior variance $\text{Var}(p_{t,c}) \rightarrow 0$ and the expected posterior after one additional observation converges to the current posterior. Hence $\mathbb{E}[\text{EU}^*(b \mid x)] \rightarrow \text{EU}^*(b)$ and $\text{VOI}(t \mid b) \rightarrow 0$. $\square$

This ensures that the agent’s querying naturally terminates: after sufficiently many observations, no tool query is worth its cost.

Myopic versus horizon-aware VOI. The VOI of eq. (6) is *myopic*: it values tool t ’s response only for the current question’s decision. An agent maximising it is a Bayesian *learner*—it updates beliefs from every response it sees—but not a Bayesian *explorer*: it never pays to query a tool whose response it will not submit, since under eq. (6) a query that cannot change the current submission has zero value. When the same category recurs across a question stream, this *under-explores*—the agent never pays to discover an expensive-but-reliable tool whose value would be amortised over future questions.

The fix is to value information over the *horizon*. Suppose a category recurs over h remaining questions. A probe of tool t now (paying r_t , observing x) yields an updated belief $b \mid x$ that improves the best single-submit decision on each of the remaining $h - 1$ questions. The horizon-aware value of the probe is

$$\text{VOI}_h(t \mid b) = (h - 1) (\mathbb{E}_{x \sim p_t}[\text{EU}^*(b \mid x)] - \text{EU}^*(b)) - r_t, \quad (8)$$

and the agent probes the tool with the highest positive VOI_h , then submits the answer its accumulated beliefs best support. This is the *decoupled* formulation: a probe is paid for its information alone, while the submission uses everything learned—which is exactly how a deployed agent that queries tools and then commits to its best estimate behaves. Equation (8) is the depth-1 lookahead; the exact reference is finite-horizon backward induction over the belief-MDP, whose gain a depth-1 policy already captures in our benchmark (§5). Crucially, this remains pure expected-utility maximisation: exploration is *emergent* from the horizon, never injected as an ε -greedy rule or an exploration bonus.

Table 1: Tool reliability by question category. All tools respond to every question (no coverage mechanism). Reliability values represent $P(\text{correct} \mid \text{category})$; chance on 4-choice questions is 0.25.

Tool	$P(\text{correct})$					Cost
	Fact.	Num.	Rec.	Misc.	Reas.	
A (Web Search)	0.70	0.20	0.65	0.25	0.40	1
B (Knowledge Base)	0.92	0.40	0.55	0.88	0.45	2
C (Calculator)	0.25	1.00	0.25	0.25	0.25	1
D (LLM Direct)	0.65	0.50	0.45	0.40	0.72	2

Category-specific beliefs. The agent maintains separate Beta parameters for each tool-category pair, indexed by the question’s category. This lets it learn, for example, that the calculator is perfectly reliable for numerical questions but uninformative for others. Crucially, the category need not be *given*: we evaluate both an *oracle* condition (categories supplied by the environment) and a *fair, inferred* condition in which a lightweight classifier predicts the category from the question text (§4.2). Whether attribution is exact or noisy turns out to govern whether exploration pays at all—the central finding of this paper.

4 Experimental Setup

4.1 Simulated Environment

We evaluate on a benchmark of 50 multiple-choice questions spanning five categories: factual, numerical, recent events, misconceptions, and reasoning. Tools are simulated with known reliability profiles (table 1). While simulated tools limit ecological validity, they enable precise isolation of the decision-making layer’s contribution and allow all agents to see identical tool responses for a given seed.

Scoring. Correct answer: +10. Wrong answer: −5. Abstention: 0. Tool costs as in table 1. The asymmetric payoff creates a break-even submission threshold of $p_{\text{correct}} > 1/3$.

4.2 Category Inference: Oracle and Fair Conditions

Reliability is category-dependent (table 1), so a category-conditioned agent must know which category each question belongs to. We evaluate two conditions:

- **Oracle (given).** The environment supplies the true category. This is the idealised condition of the original benchmark; we retain it as a *price-of-inference reference*, not as the deployment assumption.
- **Inferred (fair).** The agent receives only the raw question text and predicts the category with a lightweight classifier. This is the condition a deployed agent actually faces.

The classifier. We use a *Gaussian Naive-Bayes* classifier on offline sentence embeddings (a frozen MiniLM encoder): a *generative*, class-conditional model that fits, for each category c , an axis-aligned Gaussian $\mathcal{N}(\mu_c, \text{diag}(\sigma_c^2))$ over the embedding dimensions and a categorical prior, predicting $\mathbb{P}(c \mid \text{text}) \propto \mathbb{P}(c) \prod_d \mathcal{N}(e_d; \mu_{c,d}, \sigma_{c,d}^2)$. It is deliberately *not* a discriminative multinomial-logistic

model: the generative form requires no iterative optimisation, is conjugate-compatible with the rest of the framework, and exposes a calibrated class posterior. On the 50-question bank its leave-one-out accuracy is 0.78 (with leak-free per-dimension standardisation), well above the 0.20 five-class chance rate but far from perfect—which is precisely what makes the fair condition informative.

Using the posterior. In the inferred condition the classifier emits a posterior $\pi = (\pi_1, \dots, \pi_C)$ over categories. The agent indexes its reliability beliefs through π rather than a one-hot category, and credits each reliability update to category c in proportion to π_c (soft credit-assignment). This is where attribution noise enters the belief substrate: a misattributed update injects evidence about one category into another’s reliability estimate. The soft rule is a tractable approximation to the proper posterior-weighted update; we make this approximation and its consequences explicit in §5.2.

LLM fairness. The LLM agents (below) are evaluated on the raw question text with no category information—the same input every agent’s classifier sees. By the data-processing inequality, the inferred posterior π is a deterministic lossy function of the text and carries no category information an LLM lacks; injecting it would merely import the classifier’s error rate. We therefore reuse the saved no-category LLM runs as the fair-condition LLM baselines (verified to reproduce the paired per-seed responses bit-exactly).

4.3 Agents Compared

The Bayesian family. Three action policies share one belief substrate—Beta(1,1) priors per tool-category pair, reliability learning by conjugate update, no LLM calls, ~200 lines of Julia. They differ only in how they act on the beliefs:

- **Myopic VOI** — the VOI rule of eq. (6) (the original CREDENCE agent): query while some tool’s single-question VOI exceeds its cost, then submit or abstain.
- **Optimism-greedy** — query the single tool of highest posterior-mean reliability for the category and submit its answer. This is optimism-under-uncertainty, a *principled* Bayesian-family member (not a heuristic foil); it learns reliability exactly as the others do and differs only in committing to its current best tool rather than computing VOI.
- **Horizon-aware VOI** — the rule of eq. (8): probe by horizon value, then submit the best-learned answer.

Each policy is run in both the oracle and inferred conditions (§4.2); we write e.g. “VOI (inferred)” for the fair-condition variant.

LLM agents. Two LLM-based agents, both using native tool-calling (structured API, not text parsing). Both receive the same system prompt describing the four tools, scoring rules, the option to abstain, and a rolling history of the last 10 outcomes.

- **Claude Haiku 4.5** (claude-haiku-4-5-20251001): a production commercial model—Anthropic’s fast, low-cost tier (larger models such as Claude Sonnet and Opus sit above it and would score higher still). API cost: \$1/M input, \$5/M output tokens. We include it to show that *even the cheapest paid tier* comprehensively outscores the \$0 family; it is a lower bound on what a paid LLM buys, not an upper bound.

- **Llama 3.1 8B:** local model via Ollama, temperature 0. Zero API cost. Represents the zero-cost LLM baseline.

The comparison is between decision-making paradigms—Bayesian EU maximisation vs. LLM-prompted reasoning—not software frameworks.

Single-Best-Tool. Always queries Tool A (web search, cost 1) and submits its answer. Tests whether CREDENCE’s advantage is merely tool routing.

All-Tools. Queries all four tools, majority-votes. Demonstrates the cost of indiscriminate querying.

Random. Queries a random tool, submits its response. Lower bound.

Learning ablations. Two further variants isolate the belief substrate (used in §5.5): **no-VOI** queries all four tools every question and majority-votes (learning, but no information-value gating); **no-learning** freezes the priors (VOI gating, but no reliability updates).

4.4 Experimental Protocol

All results are averaged over 20 random seeds, with 50 questions per seed. Tool responses are pre-generated per seed so that all agents see identical (question, tool) outputs, ensuring a fair comparison; in the inferred condition the category classifier is likewise fixed across agents. LLM agents use native tool-calling via structured API and are evaluated on raw question text (§4.2). Tool reliabilities are fixed as in table 1 throughout. The exact horizon-aware reference policies (decoupled backward induction, depth sweep, matched-prior control) and the credit-rule de-confound are computed by separate offline scripts released with the code.

5 Results

We report the *fair* (inferred-category) condition as primary and retain the oracle (given-category) condition as a price-of-inference reference. The central result is a contingency law (§5.2); around it we establish three pillars—the Bayesian substrate earns the cost-efficient frontier (§5.3); exploration is fixable but its value is contingent on attribution quality (§5.2); and a per-category mechanism explains why (§5.4). Figure 1 previews both the law and the frontier. Throughout, the cost axis is *tool cost* per question (the budget that enters the score), not raw tool-call count.

5.1 Main Results (Fair Condition)

Table 2 establishes the substrate. Every *learning* policy buries the non-learning baselines (no-learning +3.8 $\approx$ chance-with-cost), and the Bayesian family occupies the cost-efficient region. The myopic-VOI agent dominates the free local Llama in the *product order*—lower tool cost (1.32 < 1.44) *and* higher gross reward (176.5 > 151.5, hence also higher score, +110.4 > +79.3)—making it the frugal frontier point, while optimism-greedy is the higher-scoring family member (+149.6) at a larger tool budget. Only paid Haiku (+445.5, 97.5% accuracy) scores above the family, at \$3.24 per 1000 questions; the family answers 56–64% correctly at \$0. We state the accuracy subtraction plainly—the claim is cost-efficiency, not parity.

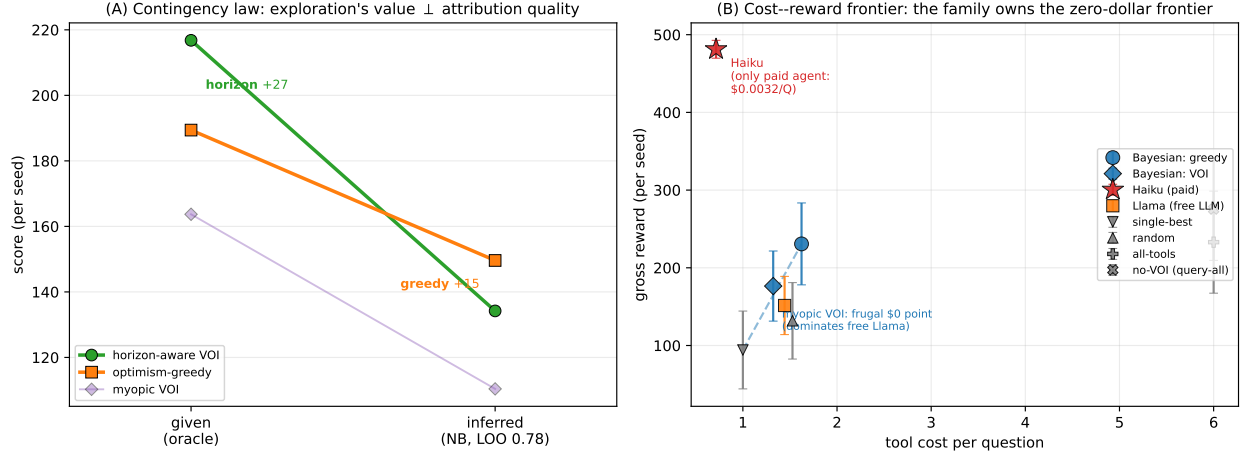


Figure 1: **(A) The contingency law.** Horizon-aware VOI beats optimism-greedy by +27 when categories are *given* (oracle) and loses by 15 once they are *inferred* (a 0.78-accuracy classifier); the crossover is the contribution. Myopic VOI (faint) loses in both conditions. **(B) The cost-reward frontier** (tool cost vs gross reward; every agent is \$0 except Haiku). The Bayesian family owns the *zero-dollar* frontier: the myopic-VOI agent is the frugal point that dominates the free local Llama (lower tool cost *and* higher gross reward), greedy is the higher-scoring family point; Haiku, the lone dollar-paying agent, sits top-left but off the \$0 frontier.

5.2 The Contingency Law

Holding the belief substrate fixed and varying only the action policy and the category condition decomposes performance into two points on one curve (table 3).

Exploration is fixable—with given categories. Myopic VOI under-explores (§5.4) and loses to greedy in both conditions (−25.7 oracle, −39.2 fair). Horizon-aware VOI fixes this: valuing information over the remaining same-category questions (pure EU maximisation—no ε -greedy, no bonus) lifts the oracle score to +216.8, **beating greedy by +27**. A *depth-1* lookahead already captures this gain (the depth sweep gives 217.6), comparable to the exact prior-optimal decoupled policy (211.8) and far below the known- θ ceiling (306.2)—so the win needs no deep search. Depth-1 slightly exceeds the prior-optimal 211.8 because the true reliabilities are extreme, which makes the Beta(1,1)-optimal policy suboptimal on them.

But the advantage is contingent on attribution quality. Under inferred categories every exploratory query injects misattribution noise into per-category reliability learning, and the advantage evaporates: horizon-aware VOI is the *most* inference-sensitive policy (−82.6). Greedy does not merely edge it out—horizon-aware VOI is Pareto-dominated in the fair condition (higher tool cost, $1.86 > 1.62$, *and* lower score, $134.2 < 149.6$), a stronger statement than the headline −15. To locate the cause, we isolate exploration—greedy’s cost-blind submission plus always-on horizon-probing, a harder-exploring setup than the deployed depth-1 agent (which is why its gaps in table 4 exceed the deployed −15)—across the credit-assignment rule; the loss is a property of imperfect attribution, not of any one rule (table 4).

Better attribution helps exploration (soft→hard, +13.7) and *hurts* minimal-query greedy (−7.3), narrowing the gap to −12 at best but never closing it; the proper posterior-weighted rule (**post**) lands between, because it lifts greedy too. Oracle attribution is the only regime in which horizon-aware

Table 2: Main results—fair (inferred-category) condition, mean over 20 paired seeds, 50 questions/seed. “Reward” is gross ($\sum +10/-5/0$, before tool cost); “Score” = reward − tool cost. “Cost/Q” is tool cost per question; “Acc.%” is the overall correct rate (abstentions count as incorrect). LLM agents are the reused no-category runs (§4.2). API \$ is per question.

Agent	Score	Reward	Acc.%	Abst.%	Cost/Q	API \$/Q
Claude Haiku 4.5 (reused)	+445.5	481.2	97.5	0.0	0.71	0.0032
Optimism-greedy	+149.6	230.8	64.1	0.0	1.62	0
Horizon-aware VOI	+134.2	227.0	63.6	0.0	1.86	0
Myopic VOI	+110.4	176.5	56.2	2.0	1.32	0
Llama 3.1 8B (reused)	+79.3	151.5	45.9	22.9	1.44	0
Random	+55.4	131.8	50.9	0.0	1.53	0
Single-Best-Tool	+44.2	94.2	45.9	0.0	1.00	0
No-learning	+3.8	53.8	40.5	0.0	1.00	0
No-VOI (query-all)	−24.5	275.5	69.3	2.3	6.00	0
All-Tools	−67.0	233.0	64.4	0.0	6.00	0

Table 3: The decomposition: score by action policy $\times$ category condition (same Beta-Bernoulli substrate throughout). The winner flips between conditions—this crossover is the contingency law. Oracle reference points: the exact prior-optimal decoupled policy scores 211.8; the deployed depth-1 horizon agent realises 216.8 (it slightly exceeds the prior-optimal policy because the true reliabilities are extreme, so the Beta(1,1)-optimal policy is not optimal on them); the only genuine ceiling is the known- θ upper bound, 306.2.

Policy	Oracle (given)	Inferred (fair)	Inference penalty
Myopic VOI	163.7	110.4	−53.3
Optimism-greedy	189.4	149.6	−39.6
Horizon-aware VOI	216.8	134.2	−82.6
Winner	horizon-VOI +27	greedy +15	—

VOI wins. **The value of exploration is contingent on category-attribution quality**—the +27 oracle result and the fair loss are two points on one curve, and the 0.78 classifier sits below the break-even.

Methods caveat: the deployed credit rule is an approximation. The deployed and headline numbers use *soft* credit-assignment—crediting each category by the classifier posterior π_c —a tractable approximation to the proper posterior-weighted update $\pi_c \cdot \text{likelihood}$ (**post** in table 4). We disclose rather than bury this: **post raises** the greedy baseline (149.8 $\rightarrow$ 151.4), and greedy still wins the fair condition under it, so the approximation does not manufacture the contingency result—if anything the proper rule favours the minimal-query baseline. Migrating the deployed agents to the proper rule is engineering follow-up.

A subordinate interaction. Table 4 also exhibits a credit-rule $\times$ query-strategy interaction: zero-leakage hard credit is best for exploration-heavy policies and worst for minimal-query greedy. This is a structural coupling, *not* a route to exploration winning—even in its best case (hard credit) horizon-aware VOI trails greedy by 12. We report it as strictly subordinate to the contingency law.

Table 4: Exploration under inferred categories, isolated across the credit-assignment rule (cost-blind submission + horizon-probing harness). Greedy wins under all three rules; better attribution narrows but never closes the gap.

Credit rule	Greedy	Horizon-VOI	Gap	
soft (π_c ; deployed)	149.8	116.7	−33.1	fractional leak to every category
hard ($\operatorname{argmax}_c \pi_c$)	142.5	130.4	− 12.1	zero leak; exploration’s best case
post (π_c ; likelihood)	151.4	126.7	−24.6	the proper posterior-weighted rule

Table 5: Per-category accuracy and net score per question (after tool cost), fair condition: myopic VOI vs optimism-greedy (★ marks the most reliable tool). Greedy wins four of five categories; VOI wins only *numerical*, where the best tool is both cheap (cost 1) and dominant (1.00-reliable).

Category	Best tool	Myopic VOI (acc / net)	Greedy (acc / net)	Δ net
factual	★KB (c2)	70.7% / +4.11	84.0% / +5.84	−1.73
misconceptions	★KB (c2)	38.6% / −0.52	57.9% / +1.99	−2.51
recent events	★web (c1)	45.6% / +1.09	55.6% / +1.98	−0.88
reasoning	★llm (c2)	42.5% / +0.12	49.0% / +0.75	−0.62
numerical	★calc (c1)	69.0% / + 4.24	60.5% / +2.48	+ 1.76

5.3 The Cost-Efficient Frontier

Pillar (i) holds in both conditions: reliability learning plus category inference buys the cost-efficient position. The product-order domination of the free Llama by the myopic-VOI agent is significant under a paired-seed bootstrap (95% CI, $B = 10,000$ resamples): greedy–Llama +70.2 [+49.8, +91.2], myopic-VOI–single-best +66.2 [+44.4, +88.1], myopic-VOI–Llama +31.1 [+6.2, +56.2] (the cost-axis ordering $1.32 < 1.44$ that completes the product-order domination is by point estimate).

The cost-scalarisation honesty. Domination depends on how the two cost axes (dollars, tool cost) are traded. Under a scalarised cost $\$ + p \cdot (\text{tool cost})$, the myopic-VOI agent is undominated only in a narrow band $p \approx 0.001\text{--}0.005$ (dollars per unit tool cost): below it greedy dominates, above it Haiku. So the *scalarised*-frontier claim is made at the family level (via greedy), while the myopic-VOI agent’s robust claim is the product-order domination of Llama; Haiku is the paid-frontier point. We make both claims and overclaim neither.

The accuracy paradox (secondary). Among non-LLM agents, higher accuracy coexists with lower net value: the query-all agents (no-VOI, all-tools) reach the highest non-LLM accuracy (69–64%) but the worst scores (−24.5, −67.0), because a tool budget of 6.0 per question destroys the margin. Maximising $\sum_q [v \cdot \mathbf{1}(\text{correct}) + v_{\text{wrong}} \cdot \mathbf{1}(\text{wrong})] - \sum_q \text{cost}_q$ is not the same as maximising its first sum when costs are non-zero.

5.4 Why Myopic VOI Under-Explores

The myopic VOI of eq. (6) values information only for the current question. Under Beta(1,1) priors every tool’s net VOI is $2.5 - \text{cost}$, so the agent economises onto cheap tools and *never pays to discover* an expensive-but-reliable tool whose value would amortise over future same-category questions (its knowledge-base query rate on KB-best categories crawls from 4% to 31% over a seed, versus greedy’s optimistic $25\% \rightarrow 66\%$). It therefore wins only where the best tool is both cheap and dominant

Table 6: Oracle skyline (given categories), 20 seeds—the price-of-inference reference for table 2. “Cost/Q” is tool cost per question.

Agent (oracle)	Score	Acc.%	Cost/Q
Horizon-aware VOI	+ 216.8	74.0	1.76
Optimism-greedy	+189.4	68.6	1.50
Myopic VOI	+163.7	62.0	1.25
No abstention	+156.9	62.6	1.25
No VOI (query-all)	+38.2	77.9	6.00
No learning	+3.8	40.5	1.00

by a wide margin—*numerical*, where the cost-1 calculator is 1.00-reliable (table 5); greedy wins the other four categories. This is exactly the under-exploration that horizon-aware VOI fixes (§5.2), and the aggregate is its mix-weighted projection: VOI overtakes greedy only above a $\sim 40\%$ numerical share, versus the benchmark’s 20%.

5.5 Ablations and the Oracle Skyline

Table 6 gives the oracle skyline—the price-of-inference reference for the fair results of table 2. Reliability learning is the dominant component: freezing it collapses the agent to chance-with-cost (+3.8). Removing VOI (querying all four tools) reaches the highest accuracy but the worst score—the accuracy paradox again. The optimism-greedy agent, far from being a quirk that “beats the full framework,” is simply the optimism-under-uncertainty member of the same Bayesian family; the substantive question this paper answers is not greedy-versus-VOI in the abstract but *under what conditions* each wins. Category inference, finally, is the dominant lever: it costs the myopic-VOI agent ~ 53 points and greedy ~ 40 (table 3)—both dwarfing the ≤ 27 that exploration buys *in these experiments*—so the single most valuable improvement is a better classifier, not a better action rule.

6 Discussion

When does exploration pay? The headline result is a contingency, not a verdict. With *given* categories, horizon-aware VOI is the best policy we tested—exploration, valued correctly over the horizon and kept inside pure expected-utility maximisation, beats optimism-greedy by +27. With *inferred* categories at a realistic 0.78 classifier accuracy, the same policy loses, because the per-category reliability signal that exploration exploits is exactly what attribution noise corrupts. The binding constraint on whether to explore is therefore *attribution quality*, not value-of-information in principle; and since category inference costs far more than any exploration policy can recover (§5.5), the dominant lever for a deployed agent is a better classifier, not a cleverer action rule. We conjecture—on this single 50-question simulated benchmark, so as a hypothesis for future work rather than an established law—that the same pattern holds for category-conditioned sequential decision problems with a learned context label more broadly: the value of an exploratory action is bounded by the fidelity with which its outcome can be attributed to the right context.

Why accuracy is the wrong metric. The accuracy paradox is not merely an artefact of our scoring system. Any deployment context with non-zero query costs—API fees, latency, rate limits—creates a divergence between accuracy and net value. Among non-LLM agents, the query-all agents reach the highest accuracy (69–64%) but the worst scores (−24.5, −67.0). Even for the commercial

LLM, the relevant metric is score-per-dollar: Haiku 4.5—the cheapest paid tier—outscores the best \$0 agent several-fold but at strictly positive dollar cost. The correct metric is expected utility, which accounts for both the value of correct answers and the cost of obtaining them.

Cheap design for LLM agents. Ay [2015] argues that an embodied agent’s controller need not be a universal approximator—it need only be sufficient for expressing desired behaviours given the agent’s embodiment constraints. The CREDENCE decision layer demonstrates this principle: a Julia inference engine maintaining Beta distributions and computing VOI outperforms a local LLM-based decision-maker that is orders of magnitude more complex, dominating the free local model outright at zero dollar cost. The LLM is powerful but expensive; using it for decisions that admit analytical solutions—resource allocation under uncertainty—is the opposite of cheap design.

The analogy to Ay’s exponential family policy models (Ay 2015, Proposition 1) is suggestive: just as Ay shows that policies parameterised by a small number of “feature vectors” derived from the embodiment constraints can be embodied universal approximators, the CREDENCE decision layer achieves strong performance using only the sufficient statistics of the Beta posteriors. The “feature vectors” are the tool-category reliability estimates; the “embodiment constraints” are the tools’ true (unknown) reliability profiles.

The computational asymmetry is stark: the CREDENCE decision layer processes each question in 0.016 seconds (closed-form Beta posterior updates and VOI computation), whilst the commercial LLM requires 2.67 seconds—a $167\times$ difference driven entirely by LLM inference overhead for tool selection decisions that admit analytical solutions.

Connection to dynamic staged trees. The per-tool, per-category Beta parameters in CREDENCE are structurally analogous to the stage parameters in a chain event graph [Freeman and Smith, 2011a]: each tool-category pair is a “situation” in the event tree sense, with its own Dirichlet-distributed transition probabilities. The agent’s task of learning which tools are reliable for which categories mirrors the CEG model selection problem of learning which situations share the same stage (i.e., the same transition distribution).

Limitations. The current evaluation uses simulated tools with known ground truth, which limits ecological validity, and the question bank is small (50 questions). Category inference uses a single off-the-shelf classifier at 0.78 accuracy; since attribution quality is the binding constraint we identify, the contingency law’s crossover point is specific to that accuracy, and a stronger classifier would shift it. A production commercial model (Claude Haiku 4.5, the fast low-cost tier—larger models would only widen the gap) comprehensively outscores the Bayesian family (+445.5 vs. +149.6 for the best \$0 agent), demonstrating that parametric world knowledge can substitute for principled tool selection—but at an API cost (\$3.24 per 1000 questions) and latency (2.67 s per question) that may be prohibitive at scale. The framework assumes tools return independent responses—correlated errors are not modelled—and the deployed inferred agents use the soft credit-assignment approximation (§5.2) rather than the proper posterior-weighted update. Distributional drift experiments are deferred to future work.

7 Conclusion and Future Work

We have presented CREDENCE, a Bayesian decision-theoretic framework for LLM tool selection that derives agent behaviour from five axioms rather than engineering it through prompting, and used it to ask *when* value-of-information tool selection pays. Our central finding is a contingency law:

a horizon-aware VOI policy—constitution-clean, with exploration emergent from pure expected-utility maximisation rather than an ε -greedy or bonus heuristic—beats an optimism-greedy rule by +27 when question categories are given, but loses once they are inferred by a 0.78-accuracy classifier, because category-attribution noise corrupts the per-category reliability signal exploration depends on. Independently of which action policy wins, the Bayesian substrate owns the cost-efficient frontier: it dominates a free local LLM outright and is beaten only by a paid commercial model at real dollar cost. The lesson for practice is that, under imperfect category attribution, the dominant lever is better inference, not cleverer exploration.

Several extensions follow directly. First, and most valuable per our ablation, a stronger category classifier: inference, not the action rule, is the dominant lever, and higher attribution accuracy would shift the contingency law’s crossover toward exploration. Second, migrating the deployed inferred agents from the soft credit-assignment approximation to the proper posterior-weighted update (§5.2). Third, evaluation with real (non-simulated) tools and distributional-drift experiments, to establish ecological validity and robustness to non-stationary reliability. Fourth, the full CREDENCE architecture extends beyond tool selection to a general-purpose agent framework with programs represented as DSL expressions (closed-loop policies in the sense of Sutton et al. 1999), meta-actions that modify the hypothesis space, and grammar evolution for discovering abstractions—a “strange loop” where the agent improves its own capacity for improvement, to be presented in a companion paper.

Finally, the CIRL alignment axiom (section 3.1) opens a natural connection to the preference learning literature: the agent’s beliefs about user preferences can be updated from implicit feedback (clicks, corrections, task completions) using the same conjugate machinery that tracks tool reliability. This would position CREDENCE not merely as an agent framework but as an alignment-by-construction approach, in which safety properties derive from the alignment axiom rather than being imposed as constraints—a direction the present work sets up but, exercising only Axioms 1–3, does not yet test.

Reproducibility. Code and data are available at <https://github.com/gfrmin/credence>. All experiments use fixed random seeds.

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